Student Performance Analysis & Prediction

Predicting final exam scores and flagging students at risk of failing — Python, scikit-learn

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1. Summary

A regression model predicts a student's final exam score from eleven pieces of information available before the exam. On unseen students it is accurate to within **6.7 percentage points on average** and explains **56% of the variation** in scores — against a baseline of 9.6 points and 0% for simply predicting the class average. Converted into an early-warning flag at a 60-point cut-off, it catches **21 of the 22 students** who went on to score below 50.

2. Dataset

The dataset is **simulated**, generated by the included `make_dataset.py`. It is not collected from real students, so it raises no privacy concerns, and because the underlying relationships are known they can be checked against what the model learns. The generator deliberately injects the data problems a real export would have.

Item	Detail
Rows / columns	662 rows (650 after removing duplicates), 13 columns — one row per student
Target	final_score — final exam result, 0–100
Features	study hours, attendance, previous grade, assignments submitted, absences, sleep hours, tutoring, internet access, extracurricular, parental education, gender
Score distribution	mean 61.3; 19.5% of students fall below the 50-point pass mark
Injected problems	12 duplicate rows; inconsistent spellings (yes / YES / Yes); 6 impossible attendance values (150%); 4 negative study-hour values; 113 missing values across four columns

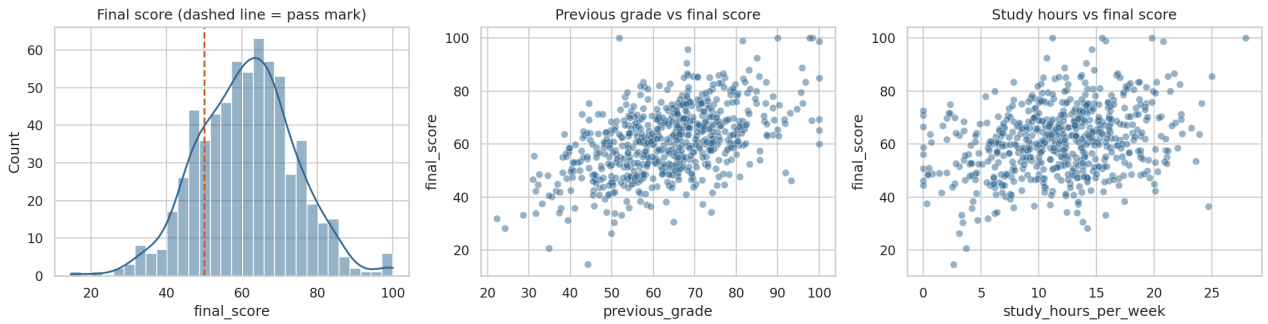


Figure 1. Final scores are roughly normal around 61. Previous grade shows the clearest relationship with the final score; study hours a weaker one.

3. Data cleaning

- **Duplicates:** 12 exact repeat rows removed (662 → 650).
- **Inconsistent categories:** yes / YES / Yes normalised to a single spelling, so one-hot encoding does not create three columns for one answer.
- **Impossible values:** 6 attendance figures above 100% and 4 negative study-hour values set to missing rather than kept or dropped — the rest of those students' data is still usable.
- **Missing values:** 113 in total. Filled inside the scikit-learn Pipeline (median for numbers, most frequent for categories) so the fill values are learned from the training rows only and nothing leaks from the test set.
- **A trap worth noting:** the parental-education category for "no formal schooling" was originally written as the word *None*, which pandas silently reads as a missing value on load. It was renamed. Without that fix, 8% of students would have been quietly imputed with the wrong category.
- **Identifier:** student_id dropped — it carries no information about performance.

4. Exploratory findings

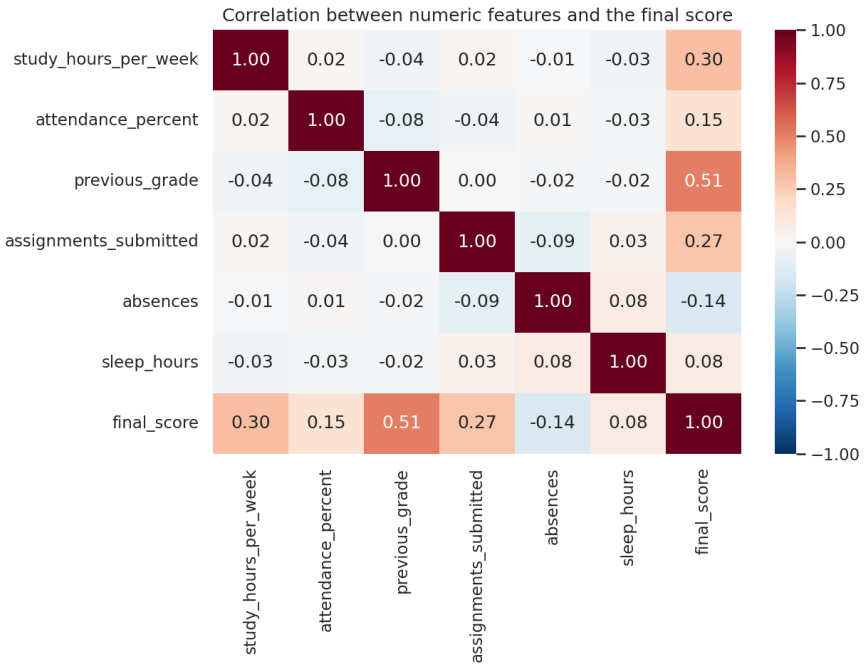


Figure 2. Correlation between the numeric features and the final score.

Previous grade is by far the strongest single predictor (r = 0.51), followed by study hours (0.30) and assignments submitted (0.27). Absences pull the other way (–0.15). Sleep hours matter least among the numeric features. Students with tutoring average 62.4 against 60.8 without — a real but small gap, and one that says nothing about cause: students who seek tutoring may already differ from those who do not.

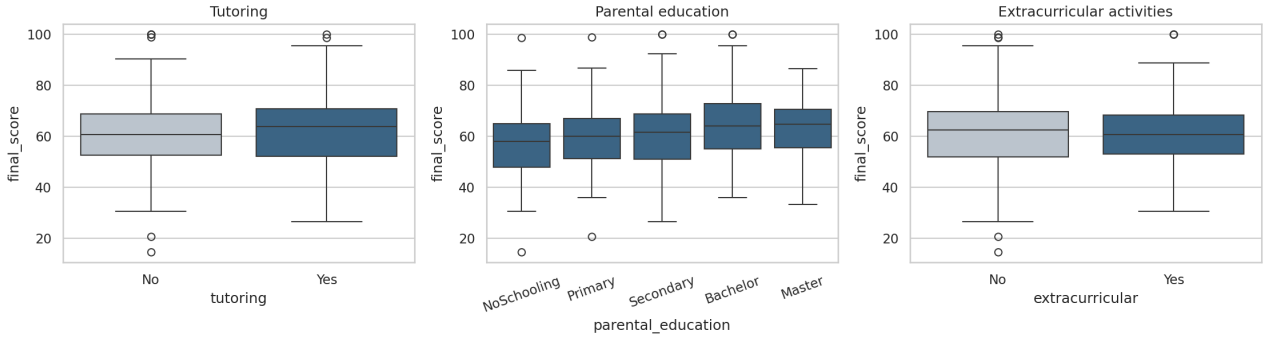


Figure 3. Score by tutoring, parental education and extracurricular activities. Parental education shows a steady upward step; the other two differences are slight.

5. Model results

Model	MAE	RMSE	R^2	Cross-validated R^2 (mean
Linear Regression	6.69	8.12	0.559	0.481 ± 0.091
Random Forest	7.03	8.76	0.487	0.416 ± 0.055
Baseline — always predict the average	9.59	12.25	–0.003	—

Measured on 130 held-out students. Selected model highlighted. MAE and RMSE are in score points, so lower is better; R^2 is the share of variation explained, so higher is better.

- **Linear Regression wins on every metric** — MAE 6.69 against 7.03, and R^2 0.559 against 0.487. Cross-validation agrees, and the gap there (0.481 vs 0.416) is larger than the variation between folds, so this is a real difference and not a lucky split.
- **Why the simpler model wins:** the relationships in this data are close to straight lines. A Random Forest spends its flexibility on curves and interactions that are not there, and fits noise instead.
- **Against the baseline:** predicting the class average for everyone gives an average error of 9.6 points. The model cuts that to 6.7 — worth having, but far from precise.
- **RMSE above MAE** (8.12 vs 6.69) means a minority of students are predicted much worse than the rest. Residuals are centred near zero (mean –0.84), so the model is not systematically over- or under-predicting.

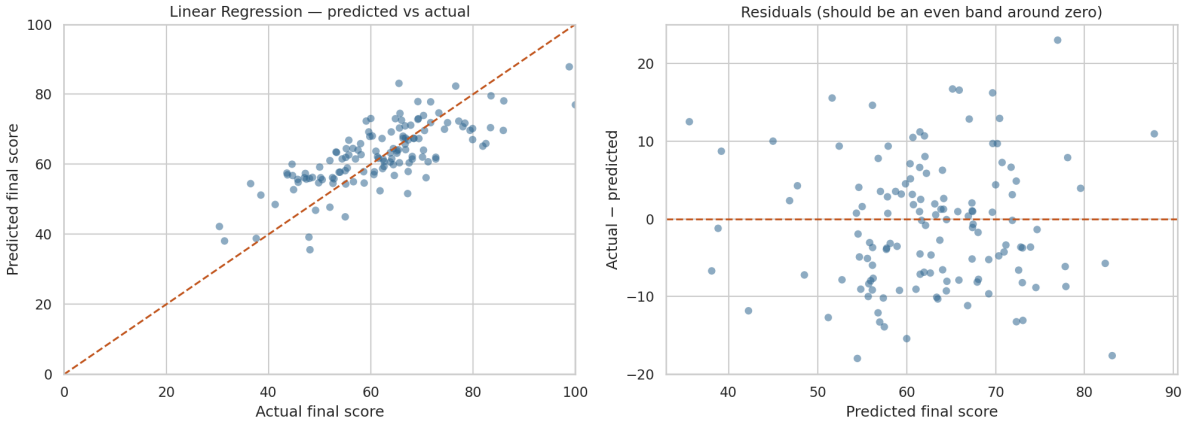


Figure 4. Left: predicted against actual, with the dashed line marking a perfect prediction. Right: residuals, which form an even band around zero — no pattern the model is missing.

6. What drives the score

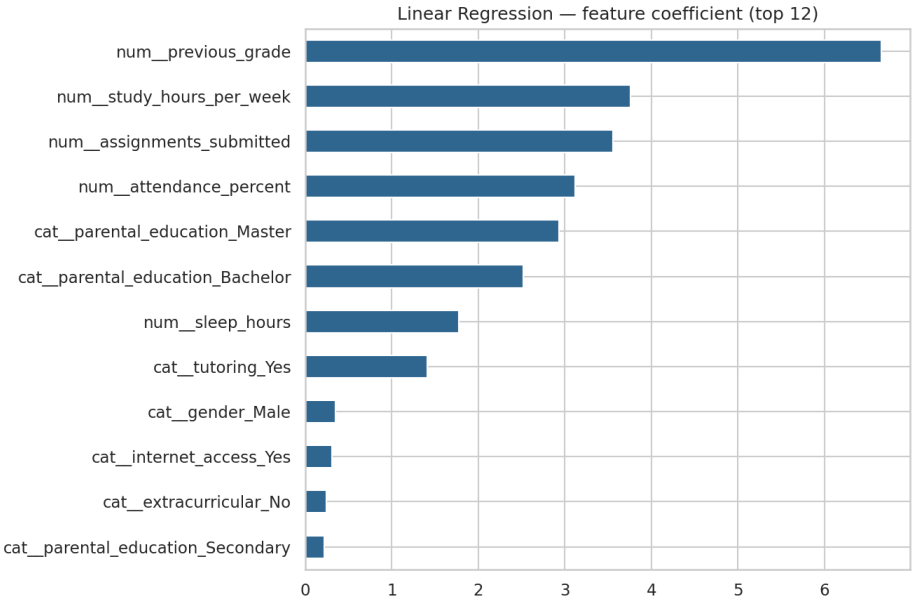


Figure 5. Linear Regression coefficients, on standardised features, so they can be compared directly.

The model recovers the pattern in the order the exploratory analysis suggested: previous grade (6.66) matters most, then study hours (3.76), assignments submitted (3.56) and attendance (3.12). Parental education at degree level adds 2.5 to 2.9 points. Tutoring adds 1.4. Reading these as levers to pull would be a mistake — they are associations in the data, not proven causes.

7. Using it as an early warning

A predicted number is not much use to a teacher on its own. Turning it into a flag is: **predicted score below a cut-off → check on this student**. Where the cut-off sits is a judgement call, not a statistical one, and the table shows what each choice costs.

Cut-off	Students flagged	Precision	Recall	What it means
50	9	0.78	0.32	Almost every flag is right, but 15 of the 22 struggling students are missed
55	19	0.63	0.55	Half caught, moderate false alarms
60	47	0.45	0.95	Catches 21 of 22 — 26 false alarms is the price
65	81	0.27	1.00	Catches everyone, but most flags are false

A cut-off of 60 is the sensible choice here. At the obvious threshold of 50 the flag misses two thirds of the students who go on to fail, which defeats the purpose. At 60 it catches 21 of 22. The false alarms are the cost: 26 students get checked on who would have passed anyway — a short conversation each, against the alternative of missing a student who needed help.

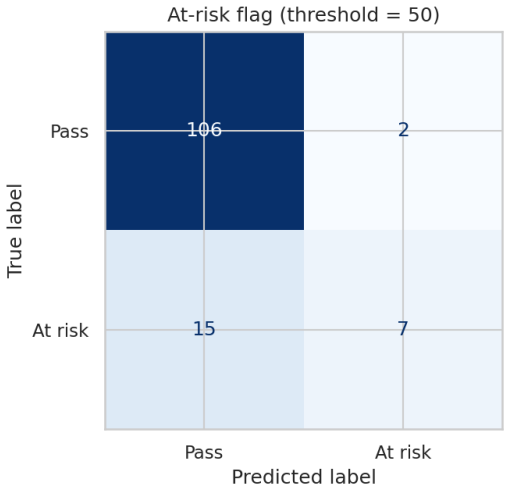


Figure 6. The flag at the default 50-point cut-off, before raising it to 60.

8. Example predictions

Study hrs	Attendance	Prev. grade	Absences	Tutoring	Predicted	Status
20	95	78	1	Yes	91.9	On track
4	58	45	18	No	26.1	AT RISK
12	80	62	5	No	62.6	On track

The third student has no recorded sleep hours; the pipeline fills the gap the same way it did during training.

9. Limitations

- **The data is simulated.** The relationships were written into the generator, so recovering them shows the pipeline works — not that these effects hold for real students. Everything here would need re-measuring on real data.
- **Correlation is not cause.** Tutoring correlates with higher scores; that does not mean assigning tutoring raises them.
- **Self-reported inputs.** Study hours and sleep are exactly the fields students over-report, and errors in the inputs become errors in the prediction.
- **Missing context.** Nothing here captures teaching quality, subject difficulty, health, paid work or family circumstances.
- **The errors are large enough to matter.** An average miss of 6.7 points, and worse for some students, means predictions near the pass mark flip easily. The flag should start a conversation, never decide anything.
- **Fairness.** Gender and parental education are inputs to the model. Before any real use, error rates must be compared across those groups — a model that is reliably worse for one group is not usable, however good the overall score. The school should consider removing those features.
- **Not production-ready.** R^2 of 0.56 is a reasonable result on simulated data, not evidence of deployment readiness. There is no external validation, no drift monitoring, and no study of what happens once staff act on the flags.

10. Conclusion

The model predicts final scores to within about 6.7 points and explains 56% of the variation, beating a class-average baseline by a clear margin. Linear Regression outperformed Random Forest on every metric, which fits data whose relationships are close to linear — the simpler model was also the better one here. Its practical value is the early-warning flag: at a 60-point cut-off it identifies 21 of the 22 students heading for a fail, early enough for someone to intervene. Used as a prompt for a conversation rather than a verdict, that is useful. Used as a judgement about a student, it is not.